

Creating MavenAnsibleWebApp1

Step 1: creates a new ansible web application project named **MavenAnsibleWebApp1** using the **maven-archetype-webapp** code.

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp1 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
```

```
samiksha@samiksha:~$ cd Devops6D
samiksha@samiksha:~/Devops6D$ mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp1 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/samiksha/Devops6D
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MavenAnsibleWebApp1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/Devops6D/MavenAnsibleWebApp1
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.140 s
[INFO] Finished at: 2026-05-11T00:18:44+05:30
[INFO]
samiksha@samiksha:~/Devops6D$ █
```

Step 2: Configure Maven Ansible Web App Project (pom.xml)

In this step, we move to the project directory using `cd MavenAnsibleWebApp` and open the Maven configuration file using `gedit pom.xml`. Then, we write the required project details, dependencies, WAR packaging settings, and build configuration for creating the web application.

```
└───┘
samiksha@samiksha:~/Devops6D$ cd MavenAnsibleWebApp1
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1$ gedit pom.xml
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/mavenv4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>

<artifactId>MavenAnsibleWebApp1</artifactId>

<packaging>war</packaging>

<version>1.0-SNAPSHOT</version>

<name>MavenAnsibleWebApp1</name>

<url>http://maven.apache.org</url>

<dependencies>

<dependency>

<groupId>junit</groupId>

<artifactId>junit</artifactId>

<version>3.8.1</version>

<scope>test</scope>

</dependency>

<!-- Servlet API for Java EE -->

<dependency>

<groupId>javax.servlet</groupId>

<artifactId>javax.servlet-api</artifactId>

<version>4.0.1</version>

<scope>provided</scope>

</dependency>
```

```
</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-war-plugin</artifactId>

<version>3.4.0</version>

<!-- Update to latest stable version -->

</plugin>

</plugins>

<finalName>MavenAnsibleWebApp1</finalName>

<!-- create a web page with the name MavenAnsibleWebApp, if we want some other name we
can chage it -->

</build>

</project>
```

Step 3: Create Ansible Configuration Files

In this step, we create a new folder named ansible inside the project directory using `mkdir ansible`. Then, we move into the folder using `cd ansible` and create the required Ansible files using `gedit` `hosts.ini` and `gedit` `playbook.yml`. These files are used to define target hosts and deployment tasks for automation.

```
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1$ mkdir ansible
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1$ cd ansible
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1/ansible$ gedit hosts.ini
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1/ansible$ gedit playbook.yml
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1/ansible$ █
```

gedit hosts.ini

This command is written inside hosts.ini file to define the Ansible inventory. It creates a group named web and adds localhost as the target system, using local connection and Python 3 interpreter for executing Ansible tasks.

```
[web] localhost ansible_connection=local
```

```
ansible_python_interpreter=/usr/bin/python3
```

gedit playbook.yml

This code is written inside playbook.yml file to automate deployment using Ansible. It targets the web host group, copies the generated WAR file from the Jenkins workspace to the Tomcat webapps folder, and then restarts the Tomcat service so the updated application runs successfully.

```
- name: Deploy to Tomcat  hosts: web  become: yes
```

```
  tasks:
```

```
- name: Copy WAR file to Tomcat
```

```
  copy:
```

```
    src:
```

```
"/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1CICD/target/MavenAnsibleWebApp1.war"
```

```
dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
```

```
remote_src: no
```

```
  - name: Stop Tomcat
```

```
    shell: /opt/tomcat/bin/shutdown.sh
```

```
    ignore_errors: yes
```

```
- name: Start Tomcat
```

shell: /opt/tomcat/bin/startup.shStep 4:

Build the Maven Ansible Web Application

In this step, we run the command `mvn clean install` to clean previous files and build the project.

```
samiksha@samiksha:~/Devops6D/MavenAnsibleWebApp1$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MavenAnsibleWebApp1 >-----
[INFO] Building MavenAnsibleWebApp1 1.0-SNAPSHOT
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MavenAnsibleWebApp1 ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/samiksha/Devops6D/MavenAnsibleWebApp1/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MavenAnsibleWebApp1 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MavenAnsibleWebApp1 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/samiksha/Devops6D/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/samiksha/Devops6D/MavenAnsibleWebApp1/src/main/webapp]
[INFO] Building war: /home/samiksha/Devops6D/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/samiksha/Devops6D/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war to /home/samiksha/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/samiksha/Devops6D/MavenAnsibleWebApp1/pom.xml to /home/samiksha/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.pom
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
```

Step 5: Verify Project Structure

In this step, we run the tree command to display the complete project structure.

```
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── Jenkinsfile
├── pom.xml
├── src
│   ├── main
│   │   ├── resources
│   │   └── webapp
│   │       ├── index.jsp
│   │       └── WEB-INF
│   │           └── web.xml
├── target
│   ├── classes
│   ├── MavenAnsibleWebApp1
│   │   ├── index.jsp
│   │   ├── META-INF
│   │   ├── WEB-INF
│   │   │   └── classes
│   │   │       └── web.xml
│   ├── MavenAnsibleWebApp1.war
│   ├── maven-archiver
│   └── pom.properties
14 directories, 10 files
```

Setting Up Jenkins Pipeline for MavenAnsibleWebApp

Step 1: To deploy **MavenAnsibleWebApp1**, open the terminal, cd into the **MavenAnsibleWebApp1** folder, create a **Jenkinsfile** containing the pipeline code, and then commit and push it to GitHub.

Edit Jenkinsfile:

```
pipeline { agent any // Use any
```

```
available agent environment {
```

```
    LANG = 'en_US.UTF-8'
```

```
    LC_ALL = 'en_US.UTF-8'
```

```
    } // this has to be added only if you get an error saying UTF required is 8 but showing in
    ISO00009
```

```

    tools {
        maven 'Maven' // Ensure this matches the name configured
in Jenkins
    }
    stages {
stage('Checkout') {
steps {
        git branch: 'master', url: 'https://github.com/shreya20m
/MavenAnsibleWebApp1CICD.git'
    }
}

    stage('Build') {
steps { sh 'mvn clean package' // Run Maven build
    }
}

    stage('Archive') {
steps { archiveArtifacts artifacts: 'target/*.war', fingerprint:true
    }
}

    stage('Deploy') {
steps { sh 'mvn clean package'
sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
    }
}
}
}
}

```

Step 2: Jenkinsfile: To deploy **MavenAnsibleWebApp1**.go to the Jenkins home page and click on “**New Item**” enter the item name and select **Pipeline** as the project type. Click “**OK**” .

New Item

Enter an item name

MavenAnsibleWebApp1-CICD

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.



Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

OK

Step 3: In **MavenAnsibleWebApp-CICD**, go to the **Pipeline** section and under **Definition**, select **Pipeline script from SCM**. Choose **Git** as the SCM, and under **Repository URL**, paste the GitHub URL of **MavenAnsibleWebApp**. In the **Credentials** field, select the previously created credential (e.g., *GitHub Credential*). Finally, in **Script Path**, enter the name of the Jenkinsfile you created in the MavenAnsibleWebApp folder and pushed to GitHub. Then click **Apply** and **Save** to finalize the configuration.

Configure

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Samiksha201104/MavenAnsibleWebApp1.git

Credentials ?

GitHub Credentials

+ Add

Advanced ▾

+ Add Repository

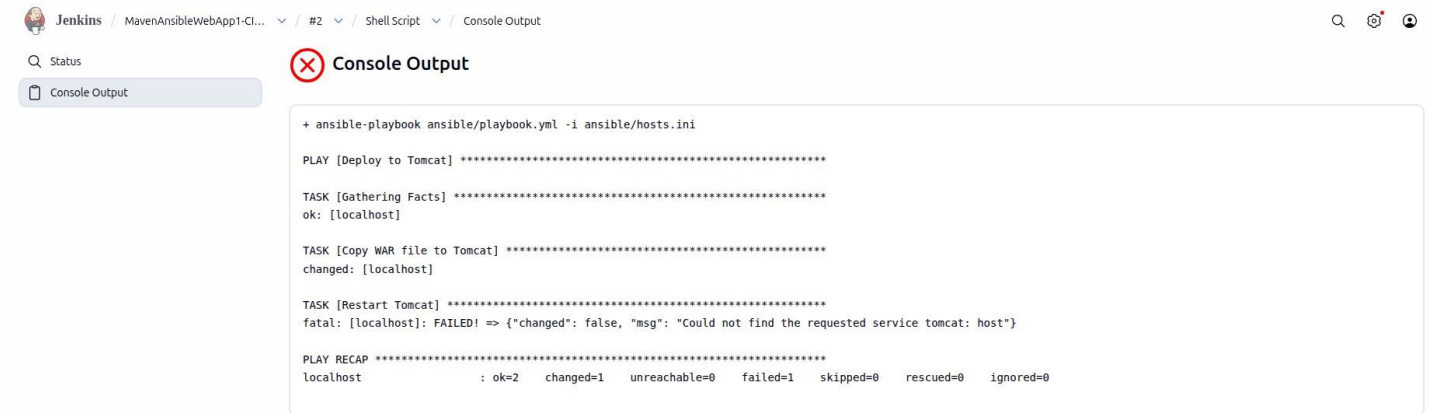
Branches to build ?

Save

Apply

Step 4: Build in Jenkins

After clicking **Apply and Save**, go back to **MavenAnsibleWebApp1-CICD** in Jenkins and click **Build Now**. And resolve the issue.



```
+ ansible-playbook ansible/playbook.yml -i ansible/hosts.ini

PLAY [Deploy to Tomcat] *****

TASK [Gathering Facts] *****
ok: [localhost]

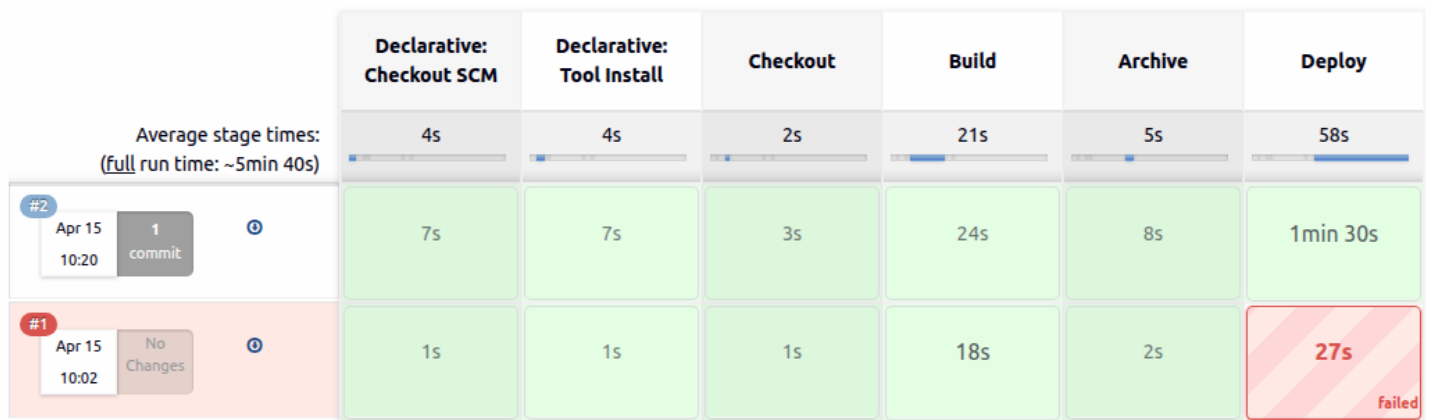
TASK [Copy WAR file to Tomcat] *****
changed: [localhost]

TASK [Restart Tomcat] *****
fatal: [localhost]: FAILED! => {"changed": false, "msg": "Could not find the requested service tomcat: host*"}

PLAY RECAP *****
localhost      : ok=2   changed=1    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

Step 5: After making the changes, go back to **MavenAnsibleWebApp1-CICD** in Jenkins and click **Build Now**. The build should complete successfully without any errors.

MavenAnsibleWebApp1-CICD - Stage View



Step 6 : Verify MavenAnsibleWebApp Deployment on Tomcat

start the server using `sh startup.sh`. After Tomcat starts successfully, we open the browser and access `http://localhost:9090` to verify the deployed application output.



✓ Password saved



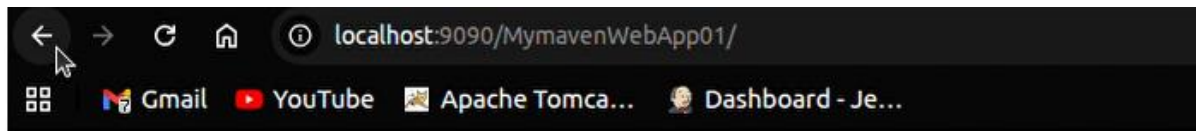
Tomcat Web Application Manager

Message: OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server S

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MavenAnsibleWebApp1	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MyMavenWebApp	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MymavenWebApp	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MymavenWebApp01	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Click on the project and to find the output.



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